

Stable training, meaningful gradients even with disjoint supports.

Stable learning of Monge maps via dual potentials

loss over Wasserstein ball **: worst-case**

#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

Learning with a Wasserstein Loss

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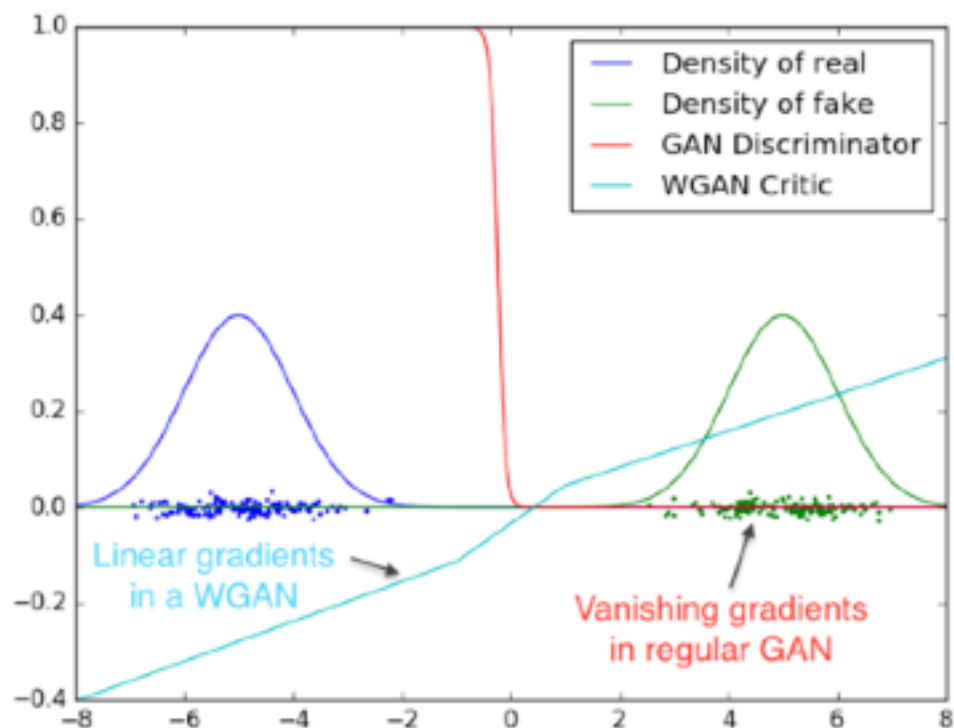
Mauricio.Araya@shell.com

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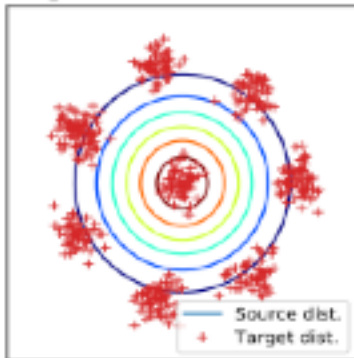
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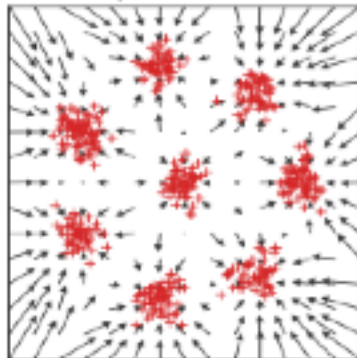
tp@ai.mit.edu



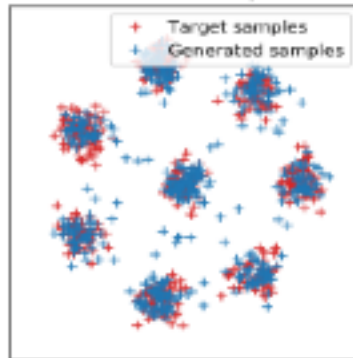
Target and Source distributions



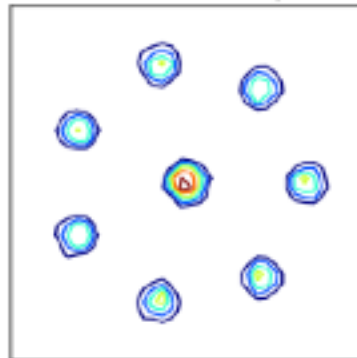
Displacement field



Generated Samples

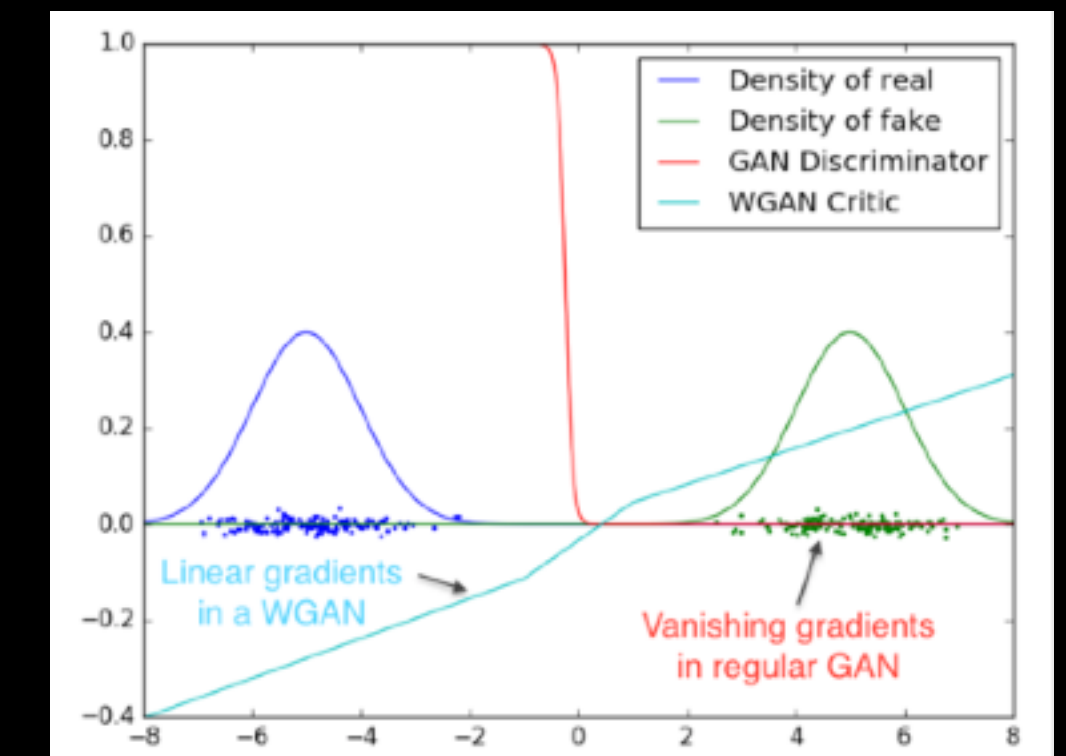


Generated density

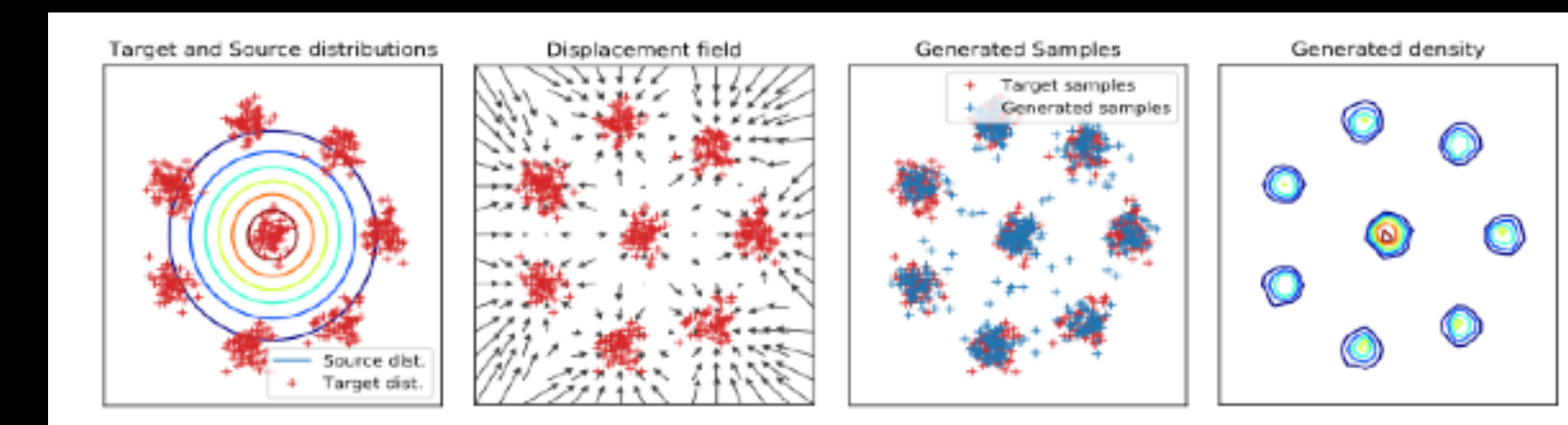


Applications in ML

- Generative Modeling
 - Wasserstein GANs (WGANs): loss is W_1 dual.
 - Stable training, meaningful gradients even with disjoint supports.
- Neural Optimal Transport
 - Stable learning of Monge maps via dual potentials.
 - Example applications: style transfer, domain adaptation
- Other uses:
 - Distributionally Robust Optimization (DRO): worst-case loss over Wasserstein ball
 - W_p as a loss for multi-label learning — — — — — — — — — — ➤
 - Duality provides tractable reformulations.



Credit: Arjovsky et al. 2017



Credit: Seguy et al. 2018

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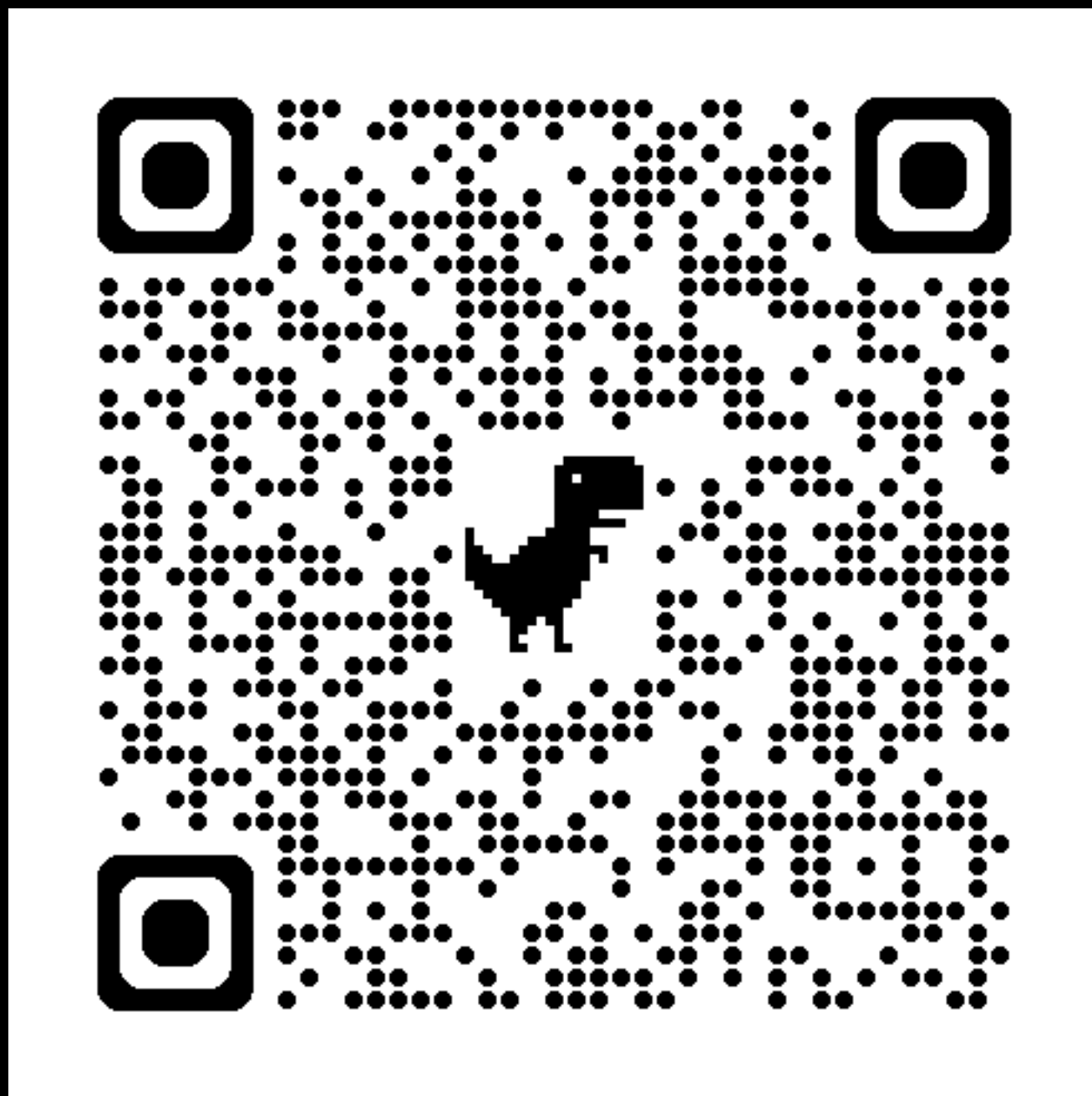
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Interactive Demo: Gelbrich / Dual

1. Compute **exact discrete** W_2 using POT (linear program).
2. Extract **dual potentials** and verify feasibility: $f_i + g_j \leq C_{ij}$
3. Implement the **Gelbrich bound** and check that $\text{Bound} \leq \text{Exact}$



<https://bit.ly/cs2840-lecture4>

Hint for 1: check documentation for `ot.emd`

Hint for 2: broadcasting is your friend (but you can also use outer products)

For 3:

$$\|\mu_\alpha - \mu_\beta\|^2 + \mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2 \leq W_2^2(\alpha, \beta)$$

$$\mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2 = \text{tr}\left(\Sigma_\alpha + \Sigma_\beta - 2\left(\Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2}\right)^{1/2}\right)$$